

Solution Requirements

Date	06 July 2026
Team ID	XXXXXX
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in or access the application to upload and analyze audio recordings. <i>(If your project has no login, write: "No authentication required for the prototype.")</i>	Medium
2	Authorization levels	Users can upload audio, view results, and download reports. Admin (if implemented) can manage reference concepts and evaluation data.	Low

3	External interfaces	Integrates with OpenAI Whisper, Sentence-BERT, Google Gemini API , and SQLite/PostgreSQL for speech processing, semantic analysis, AI feedback, and data storage.	High
		<i>services the system must connect to]</i>	
4	Transactions processing	The system accepts an audio file, transcribes speech, evaluates conceptual understanding, calculates scores, stores results, and generates a PDF report.	High
5	Reporting	Generates an evaluation report containing transcript, semantic similarity score, speech analysis, feedback, and downloadable PDF.	High
6	Business rules, etc.	Only supported audio formats (.wav, .mp3) are accepted. Evaluation is performed only after successful transcription. Scores are calculated based on semantic similarity and speech quality.	High
7	Compliance to laws or regulations	User audio and evaluation data should be handled securely and used only for educational purposes while respecting privacy principles.	Medium

8	<i>Other</i>	<i>[Add any additional functional requirements here]</i>	
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR category	Requirement Description	Target Metric/Acceptance Criteria
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1	Performance & Speed	The application should process uploaded audio and display results quickly.	Analysis completed within 10–20 seconds for a 1-minute audio file.
2	Scalability	The system should support multiple users and increasing numbers of audio files without major performance degradation.	Supports multiple analysis sessions with stable performance.
3	Security & Data Privacy	Uploaded audio, transcripts, and reports should be securely stored and protected from unauthorized access.	Secure local storage and protected API keys.

4	Reliability & Availability	The application should consistently provide analysis and report generation without failures.	99% successful processing of valid audio files.
5	Usability & Accessibility	The interface should be simple, intuitive, and easy for students and educators to use.	Streamlit UI with clear navigation and user-friendly controls.
6	<i>Other</i>	The application should be modular, easy to maintain, and executable on Windows, Linux, and macOS.	Compatible with Python 3.10+ and modular code structure.